By L. Cullen (triponacci)

Coset Index and Gauge Confinement

The arithmetic of prime splitting extends directly to the physical gauge hierarchy via the Coset Index (CI), defined as the index of the cyclic subgroup generated by the golden ratio φ:

$$\mathrm{CI} = \frac{p-1}{\mathrm{ord}(\varphi, p)} \tag{1}$$

The Standard Model gauge channels SU(3), U(1), and SU(2) correspond to the prime tower CI values $\{2, 3, 4\}$. Dynamically, the CI determines the error scaling parameter $\Upsilon = 1/\mathrm{CI}$ in the continuous error ODE. Only the SU(3) channel ($\mathrm{CI} = 2 \implies \Upsilon = 0.5$) is **supercritical** ($\Upsilon > 1/\varphi^2 \approx 0.382$), yielding an error growth root $r_1 \approx 1.781 > \varphi$. This strict supercriticality proves that strong-force confinement is an information-theoretic necessity; without confinement, the non-associative topological friction would shatter the container.

Bibliography

[1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften, vol. 322, Springer, 1999.

[2] J. S. Milne, "Class Field Theory," lecture notes, v4.03, 2020.

[3] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics, vol. 67, Springer, 1979.

[4] D. D. Wall, "Fibonacci Series Modulo m," *American Mathematical Monthly*, vol. 67, no. 6, pp. 525–532, 1960.

[5] J. W. S. Cassels and A. Fröhlich (eds.), *Algebraic Number Theory*, Academic Press, 1967.

[6] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.

[7] P. Scholze, "Perfectoid Spaces," *Publ. Math. Inst. Hautes Études Sci.*, vol. 116, pp. 245–313, 2012.